

Chapter 5: Differentiation

Section: The Derivative of a Real Function

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

Motivation

Definition of the Derivative

Differentiability Implies Continuity

Algebra of Derivatives

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Two Instructive Examples

Summary

Motivation

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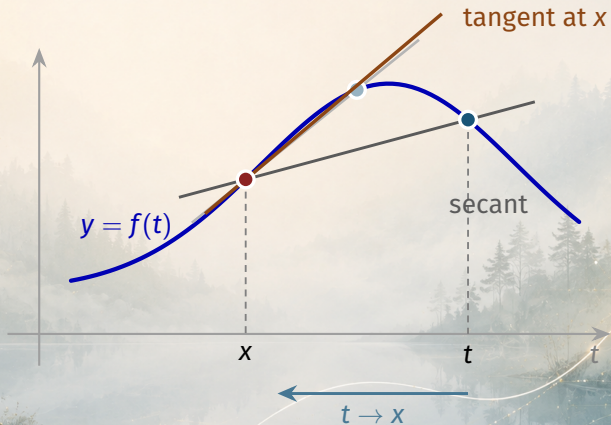
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- Preview: differentiable $\Rightarrow$ continuous, algebraic rules, the **chain rule**, and two surprising examples

The background is a serene, misty forest landscape. In the foreground, a calm lake reflects the surrounding trees and the hazy sky. The trees are mostly evergreens, with some deciduous trees showing autumnal colors. The mist is thick, creating a sense of depth and mystery. Overlaid on this natural scene are several abstract, glowing geometric elements. These include thin, white lines that form various shapes, some resembling a coordinate system or a network. There are also numerous small, golden-yellow dots scattered throughout the image, some of which are connected by lines, suggesting a data visualization or a complex mathematical structure. The overall color palette is soft and muted, with a lot of blues, greys, and earthy tones, punctuated by the warm glow of the golden dots and the light from the mist.

Definition of the Derivative

The Geometric Picture: Secants Becoming Tangents



Definition 5.1: The Derivative

Definition 5.1 (part 1: the difference quotient)

Let f be real-valued on $[a, b]$. For $x \in [a, b]$ form

$$\phi(t) = \frac{f(t) - f(x)}{t - x} \quad (a < t < b, t \neq x) \quad (1)$$

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Definition 5.1 (part 2: the derivative)

$$f'(x) = \lim_{t \rightarrow x} \phi(t), \quad (2)$$

provided this limit exists (in the sense of Definition 4.1).

Differentiable Functions

Terminology

- f' is the **derivative** of f ; its domain is the set of points x where the limit (2) exists
- If $f'(x)$ is defined, f is **differentiable at x**
- If f' is defined at every point of $E \subset [a, b]$, then f is **differentiable on E**

One-Sided Derivatives and Endpoints

Remarks on the domain

- Right-hand / left-hand limits in (2) give *one-sided* derivatives
- At the endpoints a and b , the derivative is a one-sided derivative
- If f is defined only on a segment (a, b) , then $f'(a), f'(b)$ are **not defined**



Differentiability Implies Continuity

Theorem 5.2: Differentiable $\Rightarrow$ Continuous

Theorem 5.2

Let f be defined on $[a, b]$. If f is differentiable at a point $x \in [a, b]$, then f is continuous at x .

Key idea: differentiability is a **stronger** condition than continuity.

Proof of Theorem 5.2

Proof: multiply and divide by $t - x$

As $t \rightarrow x$, by Theorem 4.4 (algebra of limits),

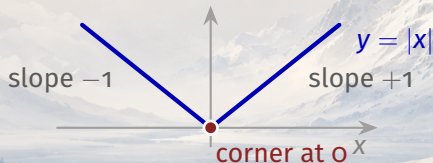
$$f(t) - f(x) = \underbrace{\frac{f(t) - f(x)}{t - x}}_{\rightarrow f'(x)} \cdot \underbrace{(t - x)}_{\rightarrow 0} \longrightarrow f'(x) \cdot 0 = 0. \quad \blacksquare$$

Technique: rewrite a difference as *quotient* $\times$ *increment*, then apply the algebra of limits.

The Converse Fails

Warning: continuity does *not* imply differentiability

- Continuous functions can fail to be differentiable at isolated points
- Chapter 7: a function continuous on **all** of $\mathbb{R}$, differentiable **nowhere**!



Algebra of Derivatives

Theorem 5.3: Sum, Product, and Quotient Rules

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Suppose f, g are defined on $[a, b]$ and differentiable at $x \in [a, b]$. Then $f + g, fg, f/g$ are differentiable at x , and

$$(a) \quad (f + g)'(x) = f'(x) + g'(x)$$

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(b) $(fg)'(x) = f'(x)g(x) + f(x)g'(x)$

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$$(a) \quad (f + g)'(x) = f'(x) + g'(x)$$

$$(b) \quad (fg)'(x) = f'(x)g(x) + f(x)g'(x)$$

$$(c) \quad \left(\frac{f}{g}\right)'(x) = \frac{g(x)f'(x) - g'(x)f(x)}{g^2(x)} \quad (\text{assuming } g(x) \neq 0)$$

Proof of Theorem 5.3(b): The Product Rule

Proof of (b): split the increment of $h = fg$

$$h(t) - h(x) = f(t)[g(t) - g(x)] + g(x)[f(t) - f(x)]$$

Divide by $t - x$; as $t \rightarrow x$, use $f(t) \rightarrow f(x)$ (Theorem 5.2):

$$\begin{aligned}\frac{h(t) - h(x)}{t - x} &= f(t) \frac{g(t) - g(x)}{t - x} + g(x) \frac{f(t) - f(x)}{t - x} \\ &\rightarrow f(x)g'(x) + g(x)f'(x). \quad \blacksquare\end{aligned}$$

Proof of Theorem 5.3(c): The Quotient Rule

Proof of (c): same idea for $h = f/g$

$$\frac{h(t) - h(x)}{t - x} = \frac{1}{g(t)g(x)} \left[g(x) \frac{f(t) - f(x)}{t - x} - f(x) \frac{g(t) - g(x)}{t - x} \right]$$

Let $t \rightarrow x$: by Theorems 4.4 and 5.2,

$$h'(x) = \frac{g(x)f'(x) - g'(x)f(x)}{g^2(x)}. \quad \blacksquare$$

Examples 5.4: Building a Library of Derivatives

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- Constants: $f(x) = c \Rightarrow f'(x) = 0$
- Identity: $f(x) = x \Rightarrow f'(x) = 1$

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- Constants: $f(x) = c \Rightarrow f'(x) = 0$
- Identity: $f(x) = x \Rightarrow f'(x) = 1$
- Repeated use of (b), (c): $(x^n)' = nx^{n-1}$ (integer n ; $x \neq 0$ if $n < 0$)
- Hence every **polynomial** is differentiable, and every **rational function** is, except where its denominator vanishes



The Chain Rule

Theorem 5.5: The Chain Rule

Theorem 5.5

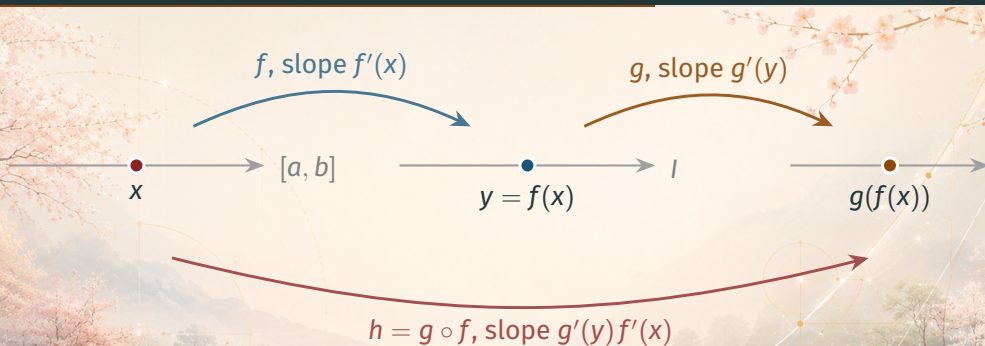
Suppose f is continuous on $[a, b]$, $f'(x)$ exists at some $x \in [a, b]$, g is defined on an interval $I \supset f([a, b])$, and g is differentiable at $f(x)$. If

$$h(t) = g(f(t)) \quad (a \leq t \leq b),$$

then h is differentiable at x , and

$$h'(x) = g'(f(x))f'(x). \quad (3)$$

Chain Rule: The Composition Picture



Intuition: small increments get scaled by $f'(x)$, then by $g'(y)$ — so the total scaling is the **product**.

Proof of Theorem 5.5

Strategy: rewrite differentiability as a **linear approximation** with a vanishing error term.

Step 1: linearize f at x and g at $y = f(x)$

$$f(t) - f(x) = (t - x) [f'(x) + u(t)], \quad (4)$$

$$g(s) - g(y) = (s - y) [g'(y) + v(s)], \quad (5)$$

where $u(t) \rightarrow 0$ as $t \rightarrow x$, and $v(s) \rightarrow 0$ as $s \rightarrow y$.

Proof of Theorem 5.5

Step 2: substitute $s = f(t)$ and chain the identities

Using first (5), then (4):

$$\begin{aligned}h(t) - h(x) &= g(f(t)) - g(f(x)) \\&= [f(t) - f(x)] \cdot [g'(y) + v(s)] \\&= (t - x) \cdot [f'(x) + u(t)] \cdot [g'(y) + v(s)]\end{aligned}$$

so for $t \neq x$:

$$\frac{h(t) - h(x)}{t - x} = [g'(y) + v(s)] \cdot [f'(x) + u(t)] \quad (6)$$

Proof of Theorem 5.5

Step 3: let $t \rightarrow x$

- f continuous at $x \Rightarrow s = f(t) \rightarrow y$
- Hence $u(t) \rightarrow 0$ and $v(s) \rightarrow 0$, so (6) gives

$$h'(x) = g'(y)f'(x) = g'(f(x))f'(x)$$



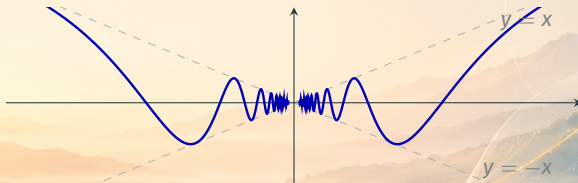


Two Instructive Examples

Example 5.6(a): $x \sin(1/x)$

Example 5.6(a)

$$f(x) = \begin{cases} x \sin \frac{1}{x} & (x \neq 0), \\ 0 & (x = 0). \end{cases} \quad (7)$$



Example 5.6(a): $x \sin(1/x)$

Away from 0: Theorems 5.3 and 5.5 apply

$$f'(x) = \sin \frac{1}{x} - \frac{1}{x} \cos \frac{1}{x} \quad (x \neq 0) \quad (8)$$

Note: at $x = 0$ these theorems say nothing — $1/x$ is not even defined.

Example 5.6(a): $x \sin(1/x)$

At $x = 0$: back to the definition

For $t \neq 0$:

$$\frac{f(t) - f(0)}{t - 0} = \sin \frac{1}{t}$$

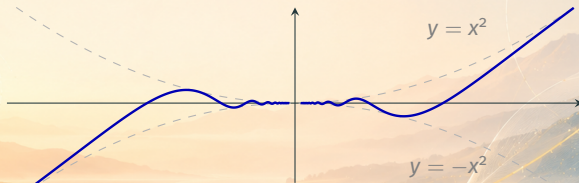
As $t \rightarrow 0$ this oscillates between -1 and 1 — **no limit exists.**

$\Rightarrow f$ is continuous at 0 , but $f'(0)$ does not exist

Example 5.6(b): $x^2 \sin(1/x)$

Example 5.6(b)

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x} & (x \neq 0), \\ 0 & (x = 0). \end{cases} \quad (9)$$



Example 5.6(b): $x^2 \sin(1/x)$

Away from 0: as before, Theorems 5.3 and 5.5 give

$$f'(x) = 2x \sin \frac{1}{x} - \cos \frac{1}{x} \quad (x \neq 0) \quad (10)$$

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At $x = 0$: the definition, plus a squeeze

$$\left| \frac{f(t) - f(0)}{t - 0} \right| = \left| t \sin \frac{1}{t} \right| \leq |t| \quad (t \neq 0) \implies f'(0) = 0 \quad (11)$$

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Punch line: f is differentiable **everywhere**, but f' is **not continuous** at 0 — the term $\cos(1/x)$ in (10) has no limit as $x \rightarrow 0$.

A serene Zen garden scene featuring a large, smooth, grey rock in the foreground on the left, surrounded by smaller rocks and patches of green moss. The ground is covered in white sand with concentric, wavy ripples. In the background, a calm body of water reflects the soft, golden light of the sky. Tall, slender bamboo stalks are visible on the right side. The entire scene is overlaid with a delicate, glowing network of thin white lines and small golden dots, creating a sense of interconnectedness and tranquility.

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Key takeaways from this section:

- **Definition 5.1:** $f'(x)$ is the limit of difference quotients — the slope of the tangent line

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Key takeaways from this section:

- **Definition 5.1:** $f'(x)$ is the limit of difference quotients — the slope of the tangent line
- **Theorems 5.2, 5.3, 5.5:** differentiable $\Rightarrow$ continuous; sum, product, quotient rules; the chain rule $h'(x) = g'(f(x))f'(x)$
- **Examples 5.6:** continuity $\nRightarrow$ differentiability; a derivative can exist everywhere yet be **discontinuous**

Coming up next: Mean Value Theorems — how the derivative controls the global behavior of f .